Supplementary Material

Nonlinear Domain Adaptation Enables Interpretable Integration of Diverse Transcriptomic Cohorts

This supplement accompanies the main text and contains Notes 1–7.

Note 1 Kernel PCA, Latent Basis Alignment, MMD, and Kernel Structure Preservation

1.1 Kernel PCA on gene vectors

We define the source domain mRNA dataset and target domain mRNA dataset as

$$X_s \in \mathbb{R}^{m\times p}, \qquad X_t \in \mathbb{R}^{n\times q}, \tag{S1.1}$$

where m and n are the numbers of samples and p and q the numbers of genes in the source and target, respectively. Each gene is described by its expression profile across the samples of its own domain. The matrices are therefore transposed and kernel PCA (kPCA) is applied across genes, so that every gene becomes a point in a low-dimensional embedding. Genes g_i^s and g_i^t are mapped to the reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ by a nonlinear function ϕ:

$$\phi : \mathcal{X} \to \mathcal{H}, \qquad g \mapsto \phi(g). \tag{S1.2}$$

Gene–gene similarities are computed through a kernel function k separately for the source and the target:

$$K^s_{ij} = k(g_i^s, g_j^s) = \langle \phi(g_i^s), \phi(g_j^s)\rangle_{\mathcal{H}}, \qquad K^t_{ij} = k(g_i^t, g_j^t) = \langle \phi(g_i^t), \phi(g_j^t)\rangle_{\mathcal{H}}. \tag{S1.3}$$

All experiments use the Gaussian (RBF) kernel $k(x, y) = \exp(-\gamma\|x - y\|^2)$ with the domain-specific bandwidth described in Section Note 3. The kernel matrix encapsulates nonlinear gene interactions and avoids computing ϕ explicitly.

The low-dimensional embedding is obtained from the eigendecomposition of the centered kernel matrices. For a domain with g genes, the centered kernel is $\tilde{K} = HKH$ with $H = I - \frac{1}{g}\mathbf{1}\mathbf{1}^\top$. Solving

$$\tilde{K}_s v_s = \lambda_s v_s, \qquad \tilde{K}_t v_t = \lambda_t v_t \tag{S1.4}$$

and keeping the top-k eigenpairs gives the embeddings (scores)

$$Z_s = (V_s)_k(\Lambda_s)_k^{1/2}, \qquad Z_t = (V_t)_k(\Lambda_t)_k^{1/2}, \tag{S1.5}$$

where V and Λ contain the top-k eigenvectors and eigenvalues. Consequently $ZZ^\top = V_k\Lambda_k V_k^\top$ is the best rank-k approximation of $\tilde{K}$ (Eckart–Young theorem [1]). The scores are centered over the genes of each domain, because the eigenvectors of a centered kernel are orthogonal to the constant vector. Unlike ordinary PCA performed across samples, this gene-wise transformation retains a coordinate for every measured gene. The eigendecomposition provides a direct, deterministic projection, while the nonlinear RBF kernel represents gene relationships without constructing the high-dimensional feature map explicitly. In all experiments $k = 2$, because the two coordinates also define the image layout (Section Note 5).

1.2 Latent basis alignment

The source and target embeddings come from two separate eigendecompositions, so their axes are defined only up to rotation and reflection, and a gene's coordinates in the two embeddings are not directly comparable. Both datasets keep their complete gene sets. Let $\mathcal{G}'$ be the set of genes measured in both datasets ($G' = |\mathcal{G}'|$), and let $A_s, A_t \in \mathbb{R}^{G'\times k}$ be the rows of Z_s and Z_t belonging to these anchor genes, in the same gene order. Orthogonal Procrustes analysis [2] finds the orthogonal matrix

$$R = \arg\min_{R^\top R = I} \|A_t R - A_s\|_F^2. \tag{S1.6}$$

Because $R^\top R = I$, $\|A_t R - A_s\|_F^2 = \|A_t\|_F^2 + \|A_s\|_F^2 - 2\,\mathrm{tr}(R^\top A_t^\top A_s)$, so the minimizer maximizes $\mathrm{tr}(R^\top M)$ with $M = A_t^\top A_s$. With the singular value decomposition $M = U\Sigma V^\top$, the maximum is attained at

$$R = UV^\top. \tag{S1.7}$$

R can contain a reflection. No translation is fitted, because the scores are centered. The same matrix is applied to all target genes, shared and unshared:

$$Z_t^{(0)} = Z_t R. \tag{S1.8}$$

The rotation preserves distances and inner products, so $Z_t^{(0)} Z_t^{(0)\top} = Z_t Z_t^\top$; the target kernel structure and the KSP loss below are unchanged by this step. The aligned embedding $Z_t^{(0)}$ is the starting point of the adaptation and defines the no-adaptation baseline (Section Note 2).

The alignment assumes that the shared genes have similar relative geometry in the two domains, and with $k = 2$ it fits one rotation angle and an optional reflection.

1.3 Maximum mean discrepancy

Given two probability distributions P and Q, the MMD is defined as

$$\mathrm{MMD}^2(P, Q; \mathcal{H}) = \|\mu_P - \mu_Q\|_{\mathcal{H}}^2, \tag{S1.9}$$

where μ_P and μ_Q are the mean embeddings of P and Q in $\mathcal{H}$ [3]. The adaptation loss is the squared MMD between the source and target embeddings, evaluated on the shared genes only,

$$\mathcal{L}_{\mathrm{MMD}} = \mathrm{MMD}^2(Z_{s,\mathrm{sh}}, Z_{t,\mathrm{sh}}), \tag{S1.10}$$

where $Z_{s,\mathrm{sh}}$ and $Z_{t,\mathrm{sh}} \in \mathbb{R}^{G' \times k}$ contain the rows of the shared genes. Expanding the squared norm as an inner product gives

$$\|\mu_{Z_s} - \mu_{Z_t}\|_{\mathcal{H}}^2 = \langle \mu_{Z_s}, \mu_{Z_s} \rangle_{\mathcal{H}} - 2\langle \mu_{Z_s}, \mu_{Z_t} \rangle_{\mathcal{H}} + \langle \mu_{Z_t}, \mu_{Z_t} \rangle_{\mathcal{H}}. \tag{S1.11}$$

The empirical mean embeddings are $\hat{\mu}_{Z_s} = \frac{1}{G'} \sum_{i=1}^{G'} \phi(Z_{s_i})$ and $\hat{\mu}_{Z_t} = \frac{1}{G'} \sum_{j=1}^{G'} \phi(Z_{t_j})$. Substituting them and using the bilinearity of the inner product,

$$\mathrm{MMD}^2(Z_s, Z_t) = \frac{1}{G'^2} \sum_{i,j} \langle \phi(Z_{s_i}), \phi(Z_{s_j}) \rangle_{\mathcal{H}} - \frac{2}{G'^2} \sum_{i,j} \langle \phi(Z_{s_i}), \phi(Z_{t_j}) \rangle_{\mathcal{H}} + \frac{1}{G'^2} \sum_{i,j} \langle \phi(Z_{t_i}), \phi(Z_{t_j}) \rangle_{\mathcal{H}}. \tag{S1.12}$$

The kernel trick, $\langle \phi(x), \phi(y) \rangle_{\mathcal{H}} = k(x, y)$, gives the final empirical estimate

$$\mathrm{MMD}^2(Z_s, Z_t) = \frac{1}{G'^2} \sum_{i,j} k(Z_{s_i}, Z_{s_j}) + \frac{1}{G'^2} \sum_{i,j} k(Z_{t_i}, Z_{t_j}) - \frac{2}{G'^2} \sum_{i,j} k(Z_{s_i}, Z_{t_j}). \tag{S1.13}$$

The first term is constant during adaptation, because the source embedding is held fixed. The complete estimate (including the first term) is used for all reported MMD values.

1.4 Kernel structure-preservation loss

Minimizing MMD alone can collapse the target representation toward the source and distort structure that is specific to the target, and it gives no signal to target genes that are absent from the source. The KSP loss penalizes deviation of the Gram matrix of the adapted target embedding from the target kernel:

$$\mathcal{L}_{\mathrm{KSP}}(Z_t) = \frac{\|\tilde{K}_t - Z_t Z_t^\top\|_F^2}{\|\tilde{K}_t\|_F^2}, \tag{S1.14}$$

where $\tilde{K}_t$ is the centered RBF kernel of the target gene vectors, built with the same bandwidth as the target kPCA (Section Note 3) and over all target genes, shared and unshared. The loss has the following properties.

- It is non-negative and is minimized, over all rank-k embeddings, by the unadapted kPCA embedding $Z_t = V_k \Lambda_k^{1/2}$ (Eckart–Young). Its minimum is generally not zero: it equals the fraction of squared kernel eigenvalues beyond the top k, $\sum_{i>k} \lambda_i^2 / \sum_i \lambda_i^2$.
- It is invariant to rotations and reflections of Z_t, so it does not prevent MMD from aligning the target to the source.
- Because $Z_t Z_t^\top$ couples every pair of target genes, it also penalizes changes to the relations between shared and unshared genes that MMD alone would introduce.

For the symmetric matrix $D = Z_t Z_t^\top - \tilde{K}_t$ and $c = \|\tilde{K}_t\|_F^2$, $d\mathcal{L}_{\mathrm{KSP}} = \frac{2}{c} \operatorname{tr}\big(D\,(dZ_t\, Z_t^\top + Z_t\, dZ_t^\top)\big) = \frac{4}{c} \operatorname{tr}\big(Z_t^\top D\, dZ_t\big)$, hence

$$\nabla_{Z_t} \mathcal{L}_{\mathrm{KSP}} = \frac{4}{\|\tilde{K}_t\|_F^2} \left(Z_t Z_t^\top - \tilde{K}_t\right) Z_t. \tag{S1.15}$$

At the kPCA embedding $\tilde{K}_t V_k = V_k \Lambda_k$, so the gradient vanishes there.

1.5 Composite objective and gradient

The target embedding is adapted by minimizing

$$\mathcal{L}(Z_t) = \mathrm{MMD}^2(Z_{s,\mathrm{sh}}, Z_{t,\mathrm{sh}}) + \lambda\, \mathcal{L}_{\mathrm{KSP}}(Z_t). \tag{S1.16}$$

Since the source–source kernel term is constant in Z_t, the gradient of the MMD term with respect to the embedding z_l of a shared target gene is

$$\nabla_{z_l} \mathrm{MMD}^2 = \frac{4\gamma_M}{G'^2} \left[\sum_j k(z_l, z_j)\,(z_j - z_l) - \sum_i k(s_i, z_l)\,(s_i - z_l) \right], \tag{S1.17}$$

where z_j are the shared target embeddings, s_i the shared source embeddings, and γ_M the MMD bandwidth. The first term describes interactions within the target domain and spreads target points, while the second models source–target interactions and attracts the target distribution toward the fixed source. Together they drive reduction of MMD. The source–source kernel term contributes no gradient because the source embedding does not change. Shared genes receive both the MMD gradient and $\lambda \nabla \mathcal{L}_{\mathrm{KSP}}$; unshared genes receive only $\lambda \nabla \mathcal{L}_{\mathrm{KSP}}$, so no gene is discarded. The KSP gradient couples shared and unshared genes through the full target Gram matrix, rather than making the unshared features independent of adaptation.

1.6 Optimization

Adaptation starts from the aligned embedding $Z_t^{(0)}$ and uses gradient descent with backtracking (Algorithm S1). Every λ is run with identical settings. The initial step size is $\eta_0 = 200$ (the MMD gradient is mean-normalized and therefore small), the step size is halved whenever a step would raise the loss, down to a floor of $10^{-4}\eta_0$, and the iteration stops when the relative change of the total loss over a 20-epoch window is at most 10^{-6}. The step size actually used, the number of epochs, and whether the stopping criterion was met are recorded for every run. A run whose loss is not finite at the step-size floor is marked as diverged and excluded from λ selection.

Algorithm S1 Adaptation with backtracking and a relative-change stopping rule.

Input: Z_s, aligned $Z_t^{(0)}$, shared-gene indices, $\tilde{K}_t$, λ, η_0, tol, window w, max_epochs
$Z \leftarrow Z_t^{(0)}, \quad \eta \leftarrow \eta_0, \quad \eta_{\min} \leftarrow 10^{-4}\eta_0$
for $e = 1, \ldots,$ max_epochs **do**
  $g \leftarrow \nabla_Z \mathcal{L}(Z)$
  repeat
    $Z' \leftarrow Z - \eta g$
    accepted $\leftarrow$ ($\mathcal{L}(Z')$ is finite **and** $\mathcal{L}(Z') \leq \mathcal{L}(Z)$) **or** $\eta \leq \eta_{\min}$
    if not accepted **then**
      $\eta \leftarrow \max(\eta/2, \eta_{\min})$
    end if
  until accepted
  if $\mathcal{L}(Z')$ is not finite **then**
    return diverged
  end if
  $Z \leftarrow Z'$
  if $e \geq w$ **and** $|\mathcal{L}_{e-w} - \mathcal{L}_e| \leq \text{tol} \cdot |\mathcal{L}_{e-w}|$ **then**
    break
  end if
end for
return Z

Note 2 Experimental Setup and Comparison Methods

2.1 Data roles

To evaluate KPCA-DA, we used a cross-domain protocol designed to simulate generalization to new target samples. For each task and transfer direction, the source dataset is used in full and the target dataset is partitioned (stratified by class) into an adaptation subset (70%) and a held-out test subset (30%). The roles of the three data sets are:

- **Source (all samples):** the source scaler, the source gene embedding, and the training of the classifier, which is trained on source images and source labels only.
- **Target adaptation subset:** used only in label-free steps: fitting the target scaler, the target gene embedding, the Procrustes rotation, the adaptation, and the selection of the kernel bandwidths and of λ. Its labels are never used.
- **Target test subset:** excluded from all adaptation and classifier fitting. Its samples are scaled with the scaler fitted on the adaptation subset, converted to images through the adaptation-set gene coordinates, and scored.

This setting ensures that the observed improvements reflect the model's ability to generalize to unseen target samples encountered post-deployment. Details of the preprocessing pipeline, including feature scaling and probe-to-gene symbol conversion for the Desmedt microarray dataset, are provided in Section Note 5.

2.2 Pipeline

The KPCA-DA pipeline proceeds in two phases: feature alignment and image-based classification. Alignment consists of a separate kPCA per domain, the Procrustes rotation of the target onto the source (Section 1.2), and the adaptation of the target embedding by MMD and KSP. We then employ DeepInsight [4] to transform the transcriptomic profiles into structured 2D image representations, where each resulting image represents a single biological sample. In this mapping, the pixel coordinates identify genes through their position in the

two-dimensional embedding, while the pixel intensity encodes the corresponding gene expression level. These images are the input of a ResNet-50 convolutional neural network [5]. The source domain images train the classifier, which is subsequently evaluated on the images generated from the 30% held-out target test subset. The classifier is not retrained or tuned on target test labels. The kernel bandwidths γ and the regularization weight λ were selected without labels, using the source dataset and the target adaptation subset only, via the procedures in Sections Note 3 and Note 4. Because the pixel coordinates of each gene are fixed once the embeddings are computed, generating an image for a new test sample only requires mapping its expression values onto the established grid. This ensures that the feature hierarchy learned by the ResNet-50 remains spatially aligned between training and deployment.

2.3 Comparison methods

The comparison methods are:

- **W/O DA (no-adaptation baseline):** the same pipeline without the adaptation step. Each dataset keeps its full gene set and its own kPCA (same k); the target embedding is rotated onto the source embedding using the shared genes, and both embeddings are placed on one pixel grid. MMD and KSP are not applied.
- **ComBat** [6]: a linear empirical-Bayes batch correction that requires identical feature sets, so non-shared genes were removed.
- **CAN** [7]: unlike the feature-space alignment used in KPCA-DA and ComBat, CAN performs contrastive alignment in the image domain.

This design allows a direct comparison between feature-space adaptation (KPCA-DA, ComBat) and latent image-domain adaptation (CAN), while the no-adaptation baseline isolates the contribution of the adaptation step itself. The ablation variant "kPCA + MMD, no KSP" corresponds to $\lambda = 0$ with all other settings unchanged. Since the KSP gradient vanishes at the starting embedding, a "KSP only" variant is identical to the no-adaptation baseline by construction.

2.4 Evaluation metrics

We report accuracy, balanced accuracy, F1, precision, sensitivity, specificity, and the area under the ROC curve.

Note 3 Kernel Bandwidth Selection: The Median Pairwise-Distance Heuristic

3.1 Overview

Two RBF bandwidths are used. The first, γ_s and γ_t, defines the kernel of the kPCA in each domain and the target kernel $\tilde{K}_t$ of the KSP loss. The second, γ_M, defines the kernel of the MMD loss. Both are set by the median heuristic and require no labels. The RBF kernel is

$$k(x, y) = \exp\left(-\gamma \cdot \|x - y\|^2\right), \tag{S3.1}$$

where $\gamma > 0$ is the bandwidth. The choice of γ determines whether the optimization produces meaningful gradient signals: if γ is too large relative to the actual distances, the kernel evaluations approach zero and all gradients vanish, preventing the optimizer from reducing the MMD regardless of the number of steps.

3.2 The median pairwise-distance heuristic

The median heuristic selects γ such that the kernel evaluates to approximately e^{-1} at the median squared pairwise distance, placing the kernel in its most informative regime, neither saturated nor near-zero.

kPCA bandwidth. For each domain, γ is computed from the gene vectors of that domain (the source samples for the source, and the adaptation subset for the target):

$$\gamma_d = \frac{1}{\text{median}\{\|g_i^d - g_j^d\|^2 : i \neq j,\ \|g_i^d - g_j^d\|^2 > 0\}}, \qquad d \in \{s, t\}. \tag{S3.2}$$

The target kernel $\tilde{K}_t$ in the KSP loss uses γ_t, so that $Z_t Z_t^\top$ is the best rank-k approximation of $\tilde{K}_t$ before adaptation.

MMD bandwidth. The heuristic is applied to the initial embeddings of the shared-gene subsets of both domains, before the optimization loop begins. Let $Z_{s,\text{sh}}$ and $Z_{t,\text{sh}}^{(0)} \in \mathbb{R}^{G' \times k}$ denote the source and the aligned target embeddings restricted to the G' shared genes. The pooled point set is

$$Z_{\text{pool}} = [\, Z_{s,\text{sh}} \,;\, Z_{t,\text{sh}}^{(0)} \,] \in \mathbb{R}^{2G' \times k}, \tag{S3.3}$$

the squared pairwise distances over distinct pairs are

$$D_{ij}^2 = \|z_{\text{pool},i} - z_{\text{pool},j}\|^2, \quad i \neq j, \tag{S3.4}$$

and the bandwidth is

$$\gamma_M = \frac{1}{\text{median}\{D_{ij}^2 : D_{ij}^2 > 0\}}. \tag{S3.5}$$

It is computed once from the whole source and the 70% target adaptation subset, prior to evaluation on the held-out target test subset, so that bandwidth selection has no access to test data. The procedure adds negligible overhead (one pairwise-distance computation) and requires no additional hyperparameter choices.

3.3 Implementation

The heuristic is implemented in `kpcada.py` as `median_heuristic_gamma` (kPCA bandwidth) and `compute_median_gamma(Zs,Zt)` (MMD bandwidth). The latter is controlled by the `auto_gamma` parameter of `fit_transform()`, `KPCA_DA`, and `tune_lambda()`. Algorithm S2 gives the procedure.

Algorithm S2 Median pairwise-distance bandwidth selection.

Input: a point set P (gene vectors of one domain, or the pooled shared-gene embeddings)
$D^2 \leftarrow \text{pdist}(P)^2$ ▷ distinct pairs
$m \leftarrow \text{median}\left(\{D_{ij}^2 : D_{ij}^2 > 0\}\right)$
$\gamma \leftarrow 1/\max(m, 10^{-8})$
return γ

Note 4 Selection of the Regularization Parameter λ

4.1 Overview

The KPCA-DA framework aligns gene expression distributions across cohorts by minimizing a composite objective in the kernel PCA embedding space. The regularization coefficient λ balances domain alignment (MMD) against preservation of the target kernel structure (KSP). This section describes the label-free selection rule.

4.2 Optimization objective

At each gradient step the following loss is minimized with respect to the adapted target embedding $Z_t \in \mathbb{R}^{q \times k}$:

$$\mathcal{L}(\lambda) = \mathcal{L}_{\text{MMD}} + \lambda \cdot \mathcal{L}_{\text{KSP}}, \tag{S4.1}$$

with $\mathcal{L}_{\mathrm{MMD}}$ and $\mathcal{L}_{\mathrm{KSP}}$ as defined in Section Note 1. Small λ prioritizes alignment, leaving the embedding free to move toward the source distribution, whereas large λ lets structure preservation dominate and constrains the degree of adaptation.

4.3 Selection procedure

Selecting λ by the lowest final MMD^2 always favors the weakest structure penalty, because KSP can only constrain the alignment. We therefore use an MMD budget rule that requires no labels. A grid $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ is evaluated, with $\lambda = 0$ always included, using the same bandwidths, initial step size, stopping rule, and aligned starting embedding for every value. Let $\mathrm{MMD}^2_{\mathrm{init}}$ be the MMD^2 of the shared genes before adaptation, $\mathrm{MMD}^2_{\mathrm{floor}}$ the lowest MMD^2 over the non-diverged runs, and

$$\mathrm{gain} = \mathrm{MMD}^2_{\mathrm{init}} - \mathrm{MMD}^2_{\mathrm{floor}}, \qquad \mathrm{budget} = \mathrm{MMD}^2_{\mathrm{floor}} + \varepsilon \cdot \mathrm{gain}. \tag{S4.2}$$

The selected value is the largest non-diverged λ with $\mathrm{MMD}^2(\lambda) \leq$ budget, that is, the strongest KSP weight that retains at least $(1 - \varepsilon)$ of the achievable MMD reduction. The default is $\varepsilon = 0.10$, and the sensitivity of the choice to ε is also reported for $\varepsilon \in \{0.05, 0.20\}$. The rule is undefined, and the run is stopped with an error, when gain ≤ 0. A warning is raised when the selected value is the largest grid value (the grid is then extended upward) or when no $\lambda > 0$ lies within the budget (the KSP term is then effectively off). The value of ε is never tuned on test accuracy.

The grid search used only the source and the 70% target adaptation subset; the held-out 30% target test subset was excluded from all selection procedures for both γ and λ, and no class labels were used.

Note 5 Datasets, Preprocessing, Image Construction, and Classifier

5.1 Cohorts

- **TCGA-BRCA** [8]: the TCGA breast invasive carcinoma cohort.
- **Desmedt et al.** [9, 10]: the TRANSBIG multicenter validation series of lymph node-negative, systemically untreated breast cancer patients profiled on Affymetrix HG-U133A arrays (GEO accession GSE7390).
- **SMC** [11, 12]: a Korean breast cancer cohort with transcriptome profiling, obtained through cBioPortal.
- **CPTAC** [13, 14]: treatment-naive primary breast cancers of the Clinical Proteomic Tumor Analysis Consortium, obtained through cBioPortal.
- **GSE45498** [15, 16]: a triple-negative breast cancer series with primary tumors and adjacent normal tissue, and lymph node metastases. Only primary tumors and their adjacent normal tissue was used for this research.

5.2 Gene vectors for the embedding

The embeddings are computed from the expression values as provided (log-scale), with each gene represented by its expression profile across the samples of its own domain. Gene filtering uses only all source samples for the source and the adaptation subset for the target. A domain-specific threshold at the 10th variance percentile removes low-variance genes, retaining approximately 90% of measured genes; the target's selected genes are used unchanged for its held-out test subset. In the general notation below, $v = 10$ is the variance-percentile cutoff (distinct from the KPCA embedding dimension $k = 2$):

$$G_{\mathrm{sel}} = \left\{ g \in G : \mathrm{Var}(g) \geq Q_v\big(\mathrm{Var}(G)\big) \right\}, \tag{S5.1}$$

where Q_v is the v-th percentile of the variance distribution over the gene set G of that domain. Filtered, unscaled expression values are used for gene-wise KPCA and for constructing the target KSP kernel; min–max scaling is used only for subsequent image intensities.

5.3 Min–max scaling of the image intensities

Before image construction, each gene is independently rescaled using `scikit-learn`'s `MinMaxScaler` with its default range $[0, 1]$. For a gene with a non-degenerate fitted range, the transformation is

$$\tilde{x}_{ig} = \frac{x_{ig} - x_{\min,g}}{x_{\max,g} - x_{\min,g}}. \tag{S5.2}$$

Constant or numerically near-constant fitted ranges are handled by the scaler's built-in safeguards; no additive ϵ is used. Each domain has its own scaler: the source scaler is fitted on all source samples, and the target scaler is fitted on the adaptation subset only. Test samples are transformed with the scaler fitted on the adaptation subset, without refitting or clipping, so their scaled values can fall outside $[0, 1]$.

5.4 Label encoding

Class labels are encoded as consecutive non-negative integers with the `scikit-learn` `LabelEncoder`. When the source and target datasets carry different label vocabularies, a single encoder is fitted on the union of both label sets, so that the integer mapping is consistent. The encoder only maps names to integers and does not use any label for fitting a model.

5.5 Probe-to-gene conversion for the Desmedt dataset

The Desmedt dataset was originally annotated with microarray probe identifiers rather than gene symbols. To enable cross-dataset gene matching, probe IDs were mapped to HUGO Gene Nomenclature Committee (HGNC) gene symbols using Ensembl BioMart, allowing feature names to be matched across the cohorts. Shared genes between datasets are identified by exact gene-symbol match.

5.6 Image construction

The two coordinates of each gene's embedding give its position on the image. One rotation fitted on the minimum bounding rectangle of the source embedding is applied to both domains, and a single minimum and maximum taken over both embeddings define the shared frame, so that an anchor gene falls on (nearly) the same pixel in both domains. The coordinates are discretized by DeepInsight binning [4] into a 224×224 pixel grid, and each sample is converted to a three-channel image whose pixel intensities are the scaled expression values of the genes at those positions. Test samples reuse the gene coordinates of the adaptation subset.

5.7 Classifier

The classifier is a ResNet-50 [5] from the `timm` library, initialized with ImageNet weights, with inputs normalized by the ImageNet mean and standard deviation. It is trained on source images and source labels only. A stratified 20% validation split of the source images selects the checkpoint with the highest validation balanced accuracy and drives the learning-rate schedule (ReduceLROnPlateau on that metric, patience 3). Training uses AdamW (learning rate 10^{-4}), batch size 16, 15 epochs, and a class-weighted cross-entropy loss, with a fixed random seed. The target test images are only scored and are never used for selection.

Note 6 Source-Fitted ICA and Distributional Distance

To visualize and quantify source–target shift in the image domain, a single independent component analysis (ICA) transform is fitted on the source images before adaptation. These reference images are flattened, pixel channels whose source standard deviation is at most 10^{-6} are discarded, and the retained entries are standardized with the reference source means and standard deviations. A whitened PCA retains up to 50 components, limited by the number of source samples and retained entries, followed by FastICA with two components and

unit-variance whitening. Both decompositions use random seed 0; FastICA allows up to 2000 iterations. The fitted pixel mask, standardization, PCA, and ICA are reused for the source and target test images both before and after adaptation. Target test images do not contribute to fitting this transform. Thus, both conditions share one coordinate system, whose component scales are set by the reference source images.

Two distances between the source ICs and the target test ICs are reported in this common space: the Wasserstein-1 distance W_1, computed as the exact optimal-transport cost (POT library) with Euclidean ground cost and uniform weights on the points, and the Euclidean distance between the two centroids. The plots show these coordinates directly, with shared axis limits. For display, equal numbers of source and target test samples are drawn using fixed sample indices across the two conditions; the distances use all source and target test samples.

Note 7 Gradient-Weighted Class Activation Mapping (Grad-CAM)

After training the ResNet-50 classifier on source domain images, Grad-CAM [17] identifies pixel regions contributing to each class score and maps these regions to genes. For the SMC→CPTAC PAM50 task reported in the main manuscript, attribution compares the no-adaptation baseline, KPCA-DA, and a source reference. The KPCA-DA classifier evaluated on all source images provides that reference.

For each class c, Grad-CAM computes the gradient of the class score y^c with respect to the feature maps A^k of the final convolutional block of the ResNet-50 (the last block of its fourth stage), globally average-pooled to obtain per-channel importance weights:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{S7.1}$$

where Z is the number of spatial locations in the feature map. The class activation map is then formed by a ReLU-gated weighted sum over all K feature channels:

$$L_{ij}^c = \mathrm{ReLU}\left(\sum_{k=1}^{K} \alpha_k^c \cdot A_{ij}^k \right). \tag{S7.2}$$

Maps are computed with the true class of each sample as the target, over the target test samples of each subtype. The sample maps returned by the Grad-CAM library are averaged to yield a single representative map L^c per subtype, without additional normalization of the mean map before gene selection. The source reference is computed in the same way from all source images with the KPCA-DA classifier.

Each gene is assigned the activation at its fixed DeepInsight pixel coordinates for that condition, using the gene identifiers retained after filtering. Multiple genes can occupy the same pixel and therefore receive the same activation. Genes are selected in two stages: first, genes whose pixel activation in the mean map satisfies $L^c \geq \tau$, with $\tau = 0.5$, are retained; second, up to 10 genes with the highest activation are kept per subtype. Ties preserve the retained gene-column order. Four subtype maps therefore yield at most 40 subtype–gene selections per condition; the number of distinct genes can be smaller because fewer genes may pass the threshold or the same gene may be selected for multiple subtypes.

For each subtype, the selected genes are compared with the 50-gene PAM50 signature [18] and with the selected genes for the corresponding source-reference subtype. An additional external validation in the manuscript uses 18 genes showing subtype-exclusive up- or down-regulation in an independent differential expression analysis of TCGA-BRCA with iPathwayGuide (Advaita Bioinformatics). These independent sets provide complementary benchmarks: overlap with established PAM50 markers and recovery of genes implicated by the external cohort. The implementation reports the number of genes passing the threshold, the selected gene names, and gene-set overlap counts. Condition-level summaries sum the overlap counts and numbers selected across subtypes; a gene selected for multiple subtypes can contribute more than once.

References

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